

DSA STUDY BLUEPRINT SERIES

109. Number of Islands Connected Components Leetcode 200

Grid Component Boundary Traversals (Number of Islands)

Section 10 • C++ Core Data Structures & Algorithms

Core Concepts & Visual Blueprint

Theoretical models and memory architectures

Key Principles

- **Problem:** Return count of connected 1s.
- Run DFS/BFS when finding a 1, sinking adjacent land elements.

Memory & Execution Blueprint

1	1	0
---	---	---

C++ Implementation Reference

Standard code layout and syntax

```
void sinkIsland(int r, int c, vector>& grid) {  
    if (r<0||c<0||r>=grid.size()||c>=grid[0].size()||grid[r][c]=='0') return;  
    grid[r][c] = '0'; // Sink land  
    sinkIsland(r+1, c, grid); sinkIsland(r-1, c, grid);  
    sinkIsland(r, c+1, grid); sinkIsland(r, c-1, grid);  
}
```

Algorithmic Complexity & Best Practices

Performance evaluations and critical guidelines

Performance Table

Aspect / Case	Complexity
Time Complexity	$O(R * C)$
Space Complexity	$O(R * C)$ stack

Key Practices & Gotchas

- **Important:** Sinking elements in-place eliminates the need to allocate extra visited matrices.
- Verify boundary conditions (e.g. empty inputs, single elements).
- Optimize dynamic memory allocations to prevent leaks.

Dry Run & Step-by-Step Execution

Trace analysis on a sample input case

Execution Walkthrough

Let's trace Dijkstra distance relaxation updates on active vertices:

Vertex popped	Adjacent edge checked	Current distance value	New path calculation	Relaxation action
U (Dist = 2)	U → V (Weight = 3)	V (Dist = 7)	New Dist = 2 + 3 = 5	Relax: Update V (Dist = 5)

Interview Variations & Optimization Pathways

Common follow-up problems and optimization strategies

Key Variations & Follow-up Questions

- **Shortest Path choices:** BFS finds unweighted short paths; Dijkstra handles positive weights; Bellman-Ford resolves negative edges.
- **Spanning Trees:** Prim's builds MST from a start node; Kruskal's sorts all edges using disjoint sets.
- **Related LeetCode Problems:** #200 Number of Islands, #743 Network Delay Time, #787 Cheapest Flights Within K Stops.